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Letter

Head-to-Tail Selective [2 + 2 + 2] Cycloadditions of Bisallenes Enabled by Ligand-Free Nickel(0)

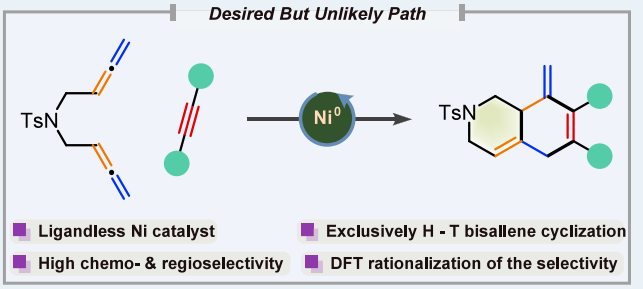
Stuart C. D. Kennington,[†] Mou Mandal,[†] Roger Monreal-Corona, Arijit Saha, Albert Poater, Anna Roglans,* and Anna Pla-Quintana*

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ABSTRACT: A ligand-free Ni(0) catalyst enables highly chemo- and regioselective [2 + 2 + 2] cycloadditions between 1,5-bisallenes and acetylenes, affording 6,6-fused hexahydroisoquinoline frameworks under mild conditions. DFT studies reveal that head-to-tail selectivity arises from competing oxidative cyclization pathways with metallacycle geometry dictating product formation. This work highlights how simple nickel complexes can achieve precise molecular construction without the need for ancillary ligands.



KEYWORDS: *1,5-bisallenes, alkynes, [2 + 2 + 2] cycloaddition, Ni(0) catalysis, hexahydroisoquinoline*

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INTRODUCTION

Controlling reaction selectivity is challenging. Transition metal catalysts[1] can guide reactants chemoselectively, and fine-tuning these catalysts to access different pathways from the same starting materials offers a powerful strategy for developing new transformations. Transition-metal-catalyzed [2 + 2 + 2] cycloadditions involving different unsaturations are invaluable tools for synthesizing both carbo- and heterocyclic complex skeletons from simple linear substrates in a single and highly atom-economical process.[2] Among the substrates, those featuring allenes,[3,4] particularly 1,5-bisallenes, have demonstrated significant promise in undergoing transition-metal-catalyzed cyclizations into structurally distinct scaffolds.[5] The versatility of the latter arises from the different ways in which the four double bonds of the two allenes can participate in the cycloaddition or cyclization process: either head-to-head, head-to-tail, tail-to-head, or tail-to-tail. Therefore, bisallenes, when reacted with a third unsaturation, can give rise to scaffolds containing five-, six-, or seven-membered rings. However, cycloisomerizations of 1,5-bisallenes, or their cyclizations with a third component, are scarce. When they occur, they typically proceed via tail-to-tail[6] or head-to-head[7] modes of cyclization in a symmetrical fashion.

Our group has already demonstrated that different final products can be obtained from the same 1,5-bisallenes by modifying the catalyst and/or the nature of the third unsaturation. When 1,5-bisallenes were reacted with dimethyl acetylenedicarboxylate (DMAD) under rhodium catalysis using two different types of ligands, a bisphosphine[8] or an N-heterocyclic carbene (NHC),[9] the cyclization of the bisallene occurred head-to-head in both cases even though the reaction follows different pathways, yielding five-membered-ring derivatives (Figure 1a,b). Remarkably, the use of a hemilabile thioether-functionalized NHC ligand favored the [2 + 2 + 2] cycloaddition reaction involving the two internal double bonds of the bisallene[9] (Figure 1b). However, when the third unsaturation was an alkene, the mode of cyclization of the starting bisallene under a Rh/bisphosphine catalytic system switched to the tail-to-tail mode of cyclization, generating derivatives with a seven-membered ring[10] (Figure 1c,d). Our group has also reported the completely intramolecular cycloaddition of an allene−yne−allene linear substrate to deliver a cross-conjugated diene with complete head-to-head mode of cyclization.[11] Aside from our work on head-to-head Rh-catalyzed [2 + 2 + 2] cycloadditions of 1,5-bisallenes with alkynes (Figure 1b),[9] the only other reported examples of [2 + 2 + 2] cycloaddition of bisallenes, to the best of our knowledge, are by Ma et al.[12] In their studies, 1,5-bisallenes underwent Rh-catalyzed [2 + 2 + 2] cycloaddition with an external allene, proceeding through a head-to-tail cyclization and leading to 6,6-fused ring systems (Figure 1e). A related [2 + 2 + 2] cycloaddition involving 1,5-allenynes and allenes affording 6,6-fused polycyclic structures under nickel catalysis was reported by Arai et al.[13] (Figure 1f). The outcome was strongly influenced by the steric bulk of the R^1 group: with

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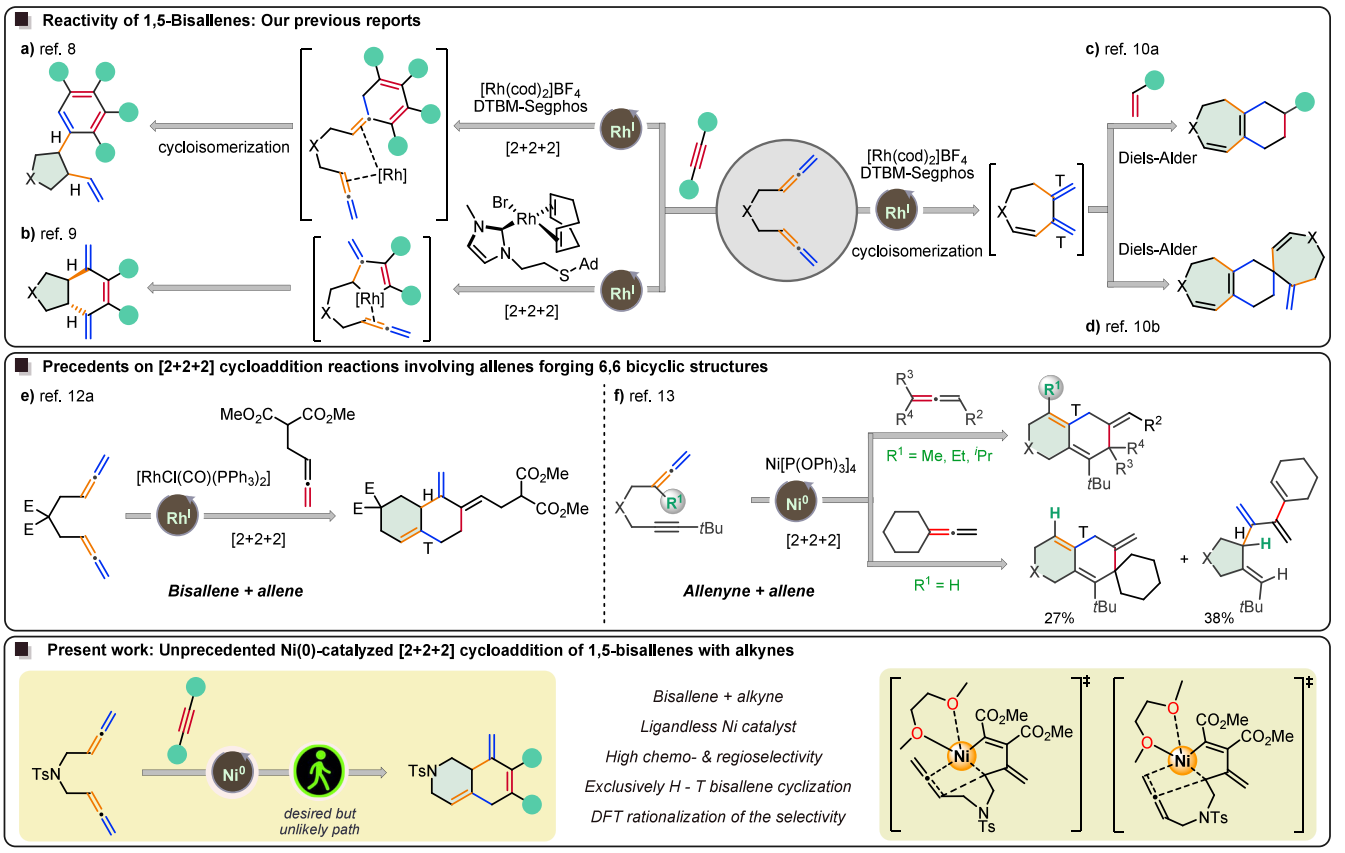


Figure 1. State-of-art of cyclization reactions of 1,5-bisallenes under transition metal catalysis.

methyl, ethyl, or isopropyl, only 6,6-fused bicyclic derivatives were obtained. In contrast, when R^1 was hydrogen, the proximal C=C bond became more reactive, giving a competing pathway leading to five-membered-ring formation (Figure 1f).

In this paper, we present the synthesis of 6,6-fused bicyclic hexahydroisoquinoline derivatives from head-to-tail cyclization of 1,5-bisallenes under nickel catalysis using alkynes as the coupling partners in the absence of any additional ligand. Additionally, a comprehensive mechanistic study using density functional theory (DFT) calculations provides insight into the origins of the observed chemo- and regioselectivity of this transformation.

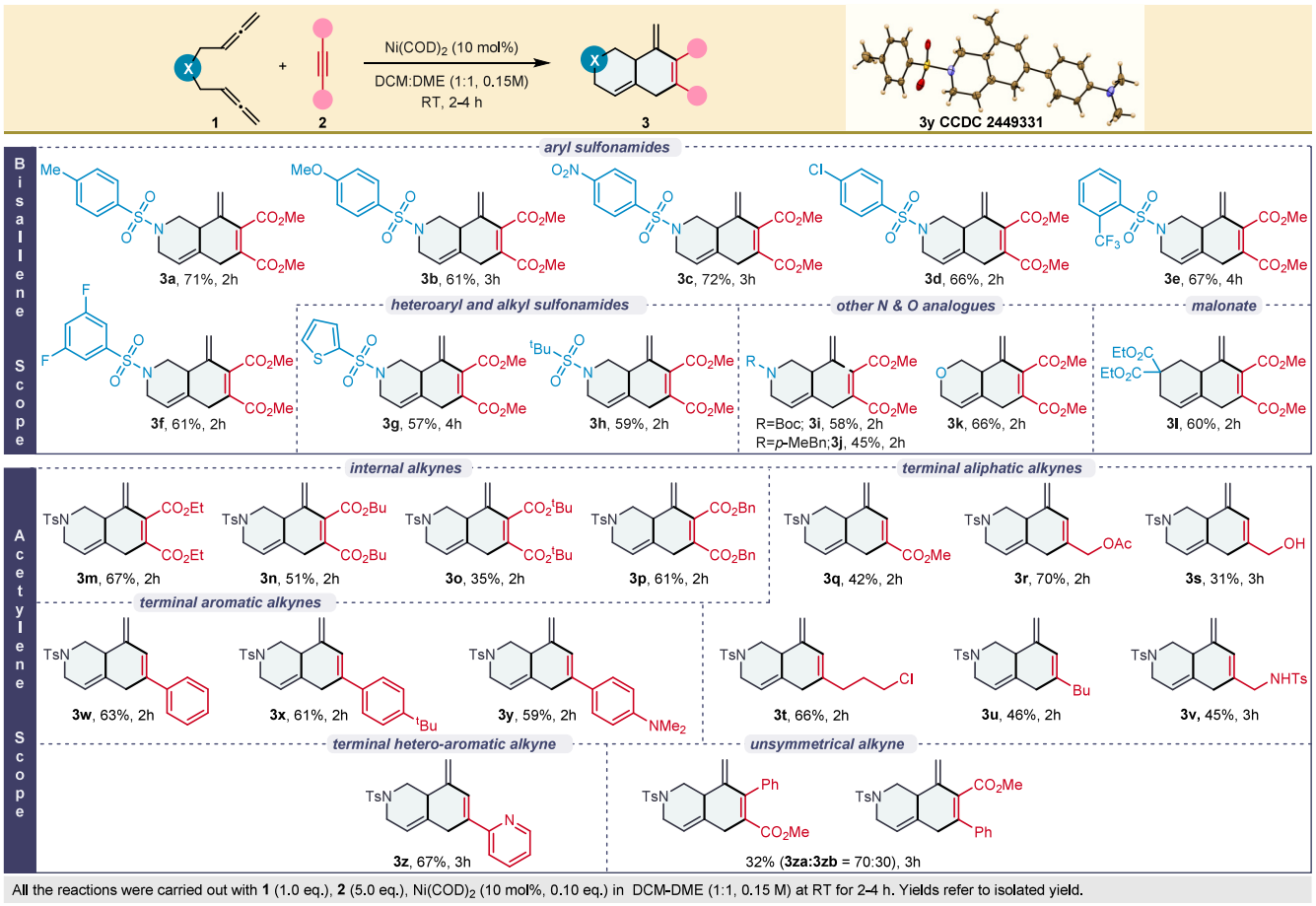
RESULTS AND DISCUSSION

We initiated our study by evaluating the reaction of *N*-tosyl-tethered 1,5-bisallene **1a** with DMAD (**2a**). Given our interest in using more abundant, relatively inexpensive, and less toxic first-row transition metal complexes, we started with ligand-free $[Ni(COD)_2]$[14] in dichloromethane (DCM) at room temperature. We observed the formation of a single product **3a** related to bisallene **1a** that incorporated DMAD and a further byproduct arising from the nickel-catalyzed trimerization of DMAD, a process already described under Ni catalysis by Coleman and Guan.[15] By NMR spectroscopy, we were able to confirm the sole formation of the 6,6-fused bicyclic product **3a** upon the highly chemoselective metallocyclization involving the head-to-tail mode of cyclization of **1a** with **2a**. After some process optimization (screening of solvents, equiv of DMAD, and Ni complexes; see Table S1), the best reaction conditions were found to be 1 equiv of bisallene **1a** and 5 equiv of DMAD using 10 mol % of $Ni(COD)_2$ in DCM/DME (1:1, 0.15 M) at room temperature. Finally, we scaled up the reaction to a 1 mmol scale for the synthesis of **3a**, which gave an identical yield of 71%, proving the scalability of the reaction.

In order to probe the generality of the reaction, we investigated the reactivity of several functionalized 1,5-bisallenes (**1a**−**1l**) as well as the compatibility of a variety of alkynes (**2a**−**2p**) under the optimized reaction conditions (Scheme 1). Aryl sulfonamide-tethered bisallenes with electron-donating groups (*p*-Me and *p*-OMe), electron-withdrawing groups (*p*-NO_2 and *o*-CF_3), and halo groups (*p*-chloro and *m*,*m*′-difluoro) on the aryl ring underwent the [2 + 2 + 2] cycloaddition reactions with excellent chemo- and regioselectivity, leading to the desired 6,6-fused bicyclic hexahydroisoquinoline derivatives **3a**−**3f** in 61−72% isolated yield. Sulfonamide tethers with a heteroaromatic ring such as thiophene (**1g**) or aliphatic substitution (^{t}butyl, **1h**) were also efficient, delivering cycloadducts **3g** and **3h** in 57% and 59% yield, respectively. With regard to the tether, not only sulfonamides but also other N-substituted tethers such as *tert*-butylcarbamate and 4-methylbenzyl bisallenes reacted with **2a**, giving compounds **3i** and **3j** in 58 and 45% yield, respectively. We then explored whether bisallenes with a tether other than nitrogen could also yield favorable reactions. An oxygen linkage could also be used in the cycloaddition, and the final product **3k** was obtained in 66% yield. Finally, diethyl malonate was tested as the tethering unit in the bisallene, providing a 60% yield of derivative **3l**. Thus, the electronic bias in 1,5-bisallene functionality had virtually no effect on the reaction outcome.

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Scheme 1. Substrate Scope of the [2 + 2 + 2] Cycloaddition of 1,5-Bisallenes and Acetylenes



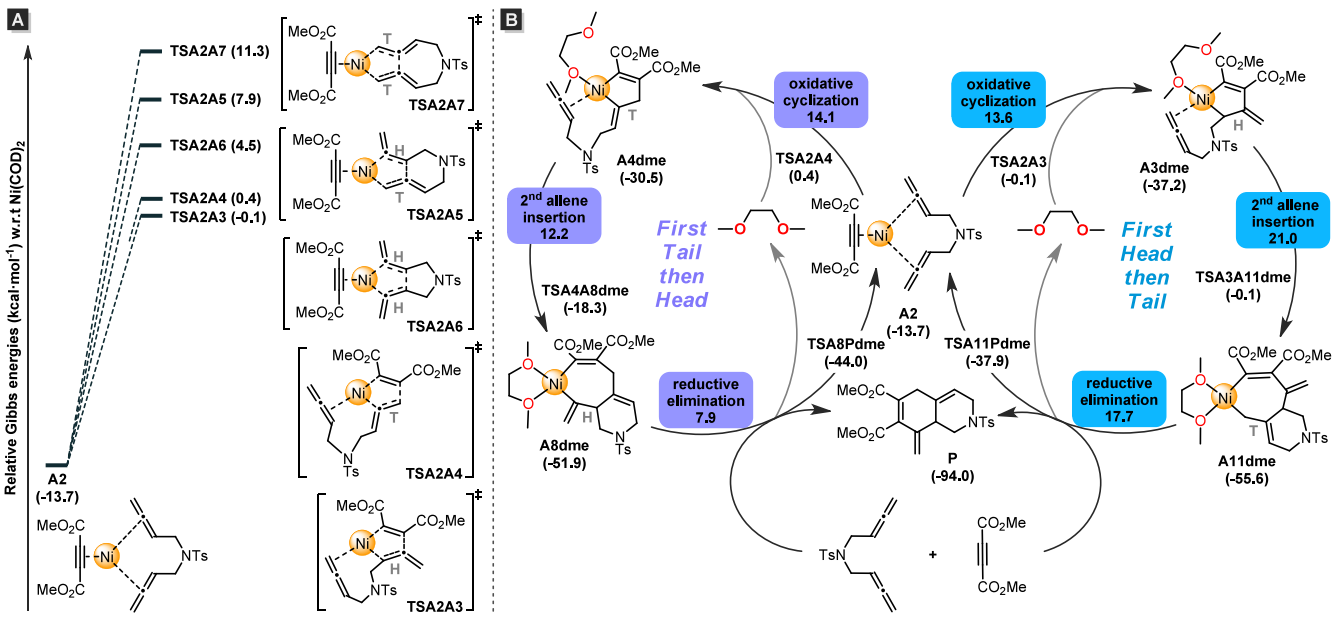


Figure 2. DFT-computed lowest-energy path for the formation of the head-to-tail product **P**.

To further illustrate the synthetic diversity, a wide array of structurally and electronically distinct alkynes were tested. The cycloaddition reaction of **1a** with other acetylenedicarboxylate esters **2b** (Et), **2c** (^{n}Bu), **2d** (^{t}Bu), and **2e** (Bn) was then carried out and delivered moderate to good yields (**3m**−**3p**, 35−67%). It is notable that the yield decreased with increasing steric bulk of the ester moiety. It is also noteworthy that the alkyne with the ethyl ester **2b** produced an inseparable mixture of the cycloadduct **3m** and the diethyl acetylenedicarboxylate trimer byproduct in a 1:0.6 ratio. Not only 1,2-disubstituted

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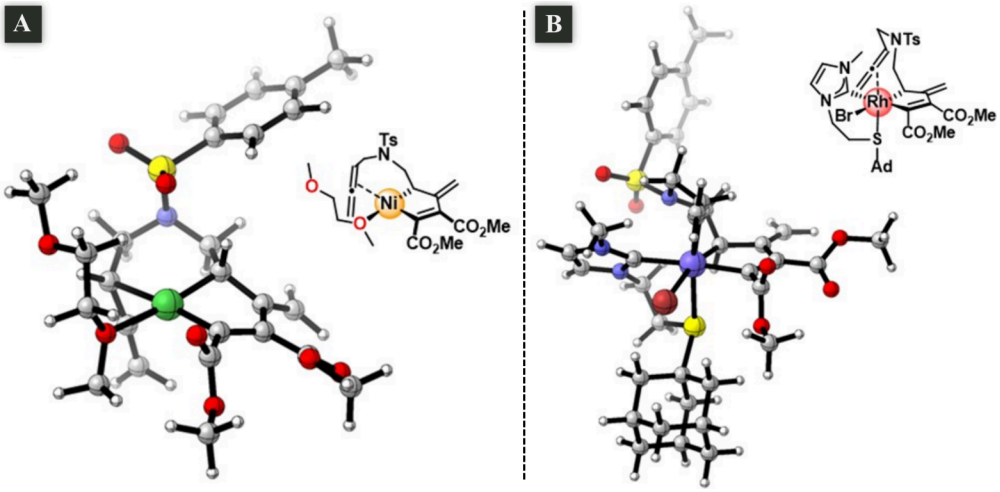


Figure 3. 3D structures of (A) square-planar Ni and (B) octahedral Rh metallacyclopentene intermediates.

alkynes but also monosubstituted alkynes (**2f**−**2k**) are relevant partners in this domino cycloaddition, unveiling exclusive regioselectivity in which the substituent is at the *β*-position with respect to the exocyclic methylidene. In this context, methyl propiolate (**2f**), propargyl acetate (**2g**), propargyl alcohol (**2h**), 5-chloro-1-pentyne (**2i**), 1-hexyne (**2j**), and tosyl-protected propargylamine (**2k**) were well-tolerated under the optimized reaction conditions, affording a range of peripherally decorated hexahydroisoquinolines **3q**−**3v** in yields between 31% and 70%. Similarly, terminal alkynes with aryl substituents such as phenylacetylene (**2l**), *p*-^{t}butylphenylacetylene (**2m**), and *p*-*N*,*N*-dimethylaminophenylacetylene (**2n**) also proved to be reactive, delivering the corresponding products **3w**−**3y** in 59−63% yield. The molecular topology of the exclusive regioisomer was further confirmed by the X-ray crystallographic analysis of compound **3y** (CCDC 2449331). The methodology was also compatible with 2-alkynylpyridine (**2o**), affording product **3z** in 67% yield without affecting the regioselectivity. Finally, the use of an unsymmetrical disubstituted acetylene (**2p**) led to an inseparable mixture of two regioisomers (**3za** and **3zb**) in a 70:30 ratio, favoring the methyl ester at the *β*-position relative to the methylidene group.

To determine the main factors favoring the observed head-to-tail selectivity, a series of DFT calculations were carried out at the B3LYP-D3/cc-pVTZ-SMD(DCM)//B3LYP-D3/cc-pVDZ level of theory (Figure 2) (see the Supporting Information for further details and for the complete set of reaction pathways studied). The first step was to determine the nature of the active catalytic species. Upon mixing Ni(COD)$_2$, the alkyne, and the bisallene in the presence of dimethoxyethane (dme)—explicitly modeled, as it is crucial for efficient ligand exchange—two main species are generated: the 16-electron species **A2** and the 18-electron species Ni(dme)(alkyne)$_2$ (see Figure S2). Optimization of these key species in both the singlet and triplet spin states revealed that the singlet state is lower in energy. Therefore, the singlet state was considered for all subsequent mechanistic studies. On the other hand, Ni(dme)(alkyne)$_2$ is unproductive for the observed transformation but accounts for the formation of minor amounts of alkyne cyclotrimerization product (see Figure S3), observed during reaction optimization. Intermediate **A2** can undergo oxidative cyclization to form five distinct intermediates, depending on the unsaturated bonds involved (see Figure S4). These include both intramolecular transformations engaging only the bisallene substrate and intermolecular processes involving the alkyne and bisallene. The three intramolecular oxidative cyclization modes (head-to-head, head-to-tail, and tail-to-tail) are energetically higher than the two intermolecular cyclizations. Therefore, oxidative cyclization involving the alkyne and the bisallene proceeds via **TSA2A4** ($\Delta G^{\ddagger}$ = 14.1 kcal·mol^{-1}), where the tail double bond of the allene reacts, or via **TSA2A3** ($\Delta G^{\ddagger}$ = 13.6 kcal·mol^{-1}), where the head double bond is involved. These results indicate that the oxidative cyclization step is not the head-to-tail selectivity-determining step. Furthermore, within the intermolecular scenario, reaction at the head double bond of the allene is only marginally favored over the tail (by 0.5 kcal·mol^{-1}), implying that both pathways could be productive. Consequently, both were further explored to understand the formation of the experimentally observed head-to-tail product. The full reaction pathways involving **TSA2A3** and **TSA2A4** are shown in Figures S5 and S6, respectively, with a schematic summary in Figure 2. In both cases, after oxidative cyclization, dme coordinates to the nickelcyclopentene intermediate, forming **A3dme** and **A4dme**. There is a slight trend between the kinetic cost and the buried volume at the Ni center (see Table S2).[16] This coordination stabilizes the intermediates and effectively renders the previous step irreversible, especially for **A4**. The subsequent allene insertion is the regioselectivity-determining step. For **A3**, the tail double bond of the second allene inserts preferentially via **TSA3A11dme** (21.0 kcal·mol^{-1}), affording the head-to-tail nickelacycle **A11dme** (−55.6 kcal·mol^{-1}). An alternative pathway leading to the head-to-head product has a higher activation barrier (29.6 kcal·mol^{-1}), 8.6 kcal·mol^{-1} above **TSA3A11dme**, and is therefore kinetically disfavored (see Figure S5). From **A4dme**, the pathway toward the head-to-tail product proceeds via **TSA4A8dme** (12.2 kcal·mol^{-1}). In contrast, formation of the unobserved tail-to-tail product requires a much higher activation barrier (26.3 kcal·mol^{-1}) and is thus not favored (see Figure S6). The reaction then proceeds via reductive elimination: from **A11dme** via **TSA11Pdme** (17.7 kcal·mol^{-1}) and from **A8dme** via **TSA8Pdme** (12.2 kcal·mol^{-1}), both delivering the same head-to-tail product **P** at −94.0 kcal·mol^{-1}. Overall, the transformation is both kinetically accessible and highly

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exergonic along both pathways. Comparing the two, oxidative cyclization is more favorable for the pathway initiated at the head double bond, although the subsequent steps are more favorable for the pathway starting at the tail. Given the small energy difference between **TSA2A3** and **TSA2A4** and the irreversible nature of the oxidative cyclization, both intermolecular pathways are considered productive. Finally, we examined the different selectivity observed in the reactions between bisallenes and alkynes catalyzed by the Rh-NHC hemilabile thioether ligand system[9] and by the ligand-free nickel catalyst. In both cases, the pathway proceeds via oxidative cyclization between the alkyne and the allene. However, the resulting metallacyclopentene intermediate has markedly different geometries: while the Rh intermediate adopts an octahedral geometry, the Ni intermediate is square-planar (Figure 3). This geometrical difference influences the preferred mode of allene insertion. Therefore, further efforts to control the selectivity in this reaction should focus on tuning the geometry of the metallacyclopentene intermediates.

SUMMARY

We have developed a ligand-free Ni(0)-catalyzed [2 + 2 + 2] cycloaddition of 1,5-bisallenes with mono- and disubstituted alkynes, affording 6,6-fused bicyclic hexahydroisoquinolines with excellent chemo- and regioselectivity under mild conditions. Notably, head-to-tail bisallene cyclization predominates, and high regioselectivity is observed for monosubstituted alkynes. Furthermore, DFT studies suggest that the observed head-to-tail selectivity arises from two competing pathways, both initiated by oxidative cyclization between the allene and the alkyne. Finally, the geometry of the metallacyclopentene intermediate plays a key role in directing the reactivity and selectivity, underscoring the importance of catalyst design in future efforts to control the regioselectivity in these transformations.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.5c04046.

Experimental procedures, characterization data, X-ray crystallographic data, and NMR spectra for all new compounds (PDF)

Cartesian coordinates (XYZ)

Crystallographic data for compound **3y** (CIF)

Accession Codes

Crystallographic data for compound **3y** (CCDC 2449331) can be downloaded free of charge from the Cambridge Crystallographic Data Centre (www.ccdc.cam.ac.uk).

AUTHOR INFORMATION

Corresponding Authors

Anna Roglans − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain;* orcid.org/0000-0002-7943-5706; Email: anna.roglans@udg.edu

Anna Pla-Quintana − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain;* orcid.org/0000-0003-2545-9048; Email: anna.plaq@udg.edu

Authors

Stuart C. D. Kennington − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain;* orcid.org/0000-0001-6804-7794

Mou Mandal − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain*

Roger Monreal-Corona − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain;* orcid.org/0000-0003-3071-9887

Arijit Saha − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain;* orcid.org/0000-0003-1354-814X

Albert Poater − *Institut de Química Computacional i Catàlisi (IQCC) and Departament de Química, Facultat de Ciències, Universitat de Girona (UdG), 17003 Girona, Catalunya, Spain;* orcid.org/0000-0002-8997-2599

Complete contact information is available at: https://pubs.acs.org/10.1021/acscatal.5c04046

Author Contributions

†S.C.D.K. and M.M. contributed equally to this work. The manuscript was written through contributions of all authors. All of the authors approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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